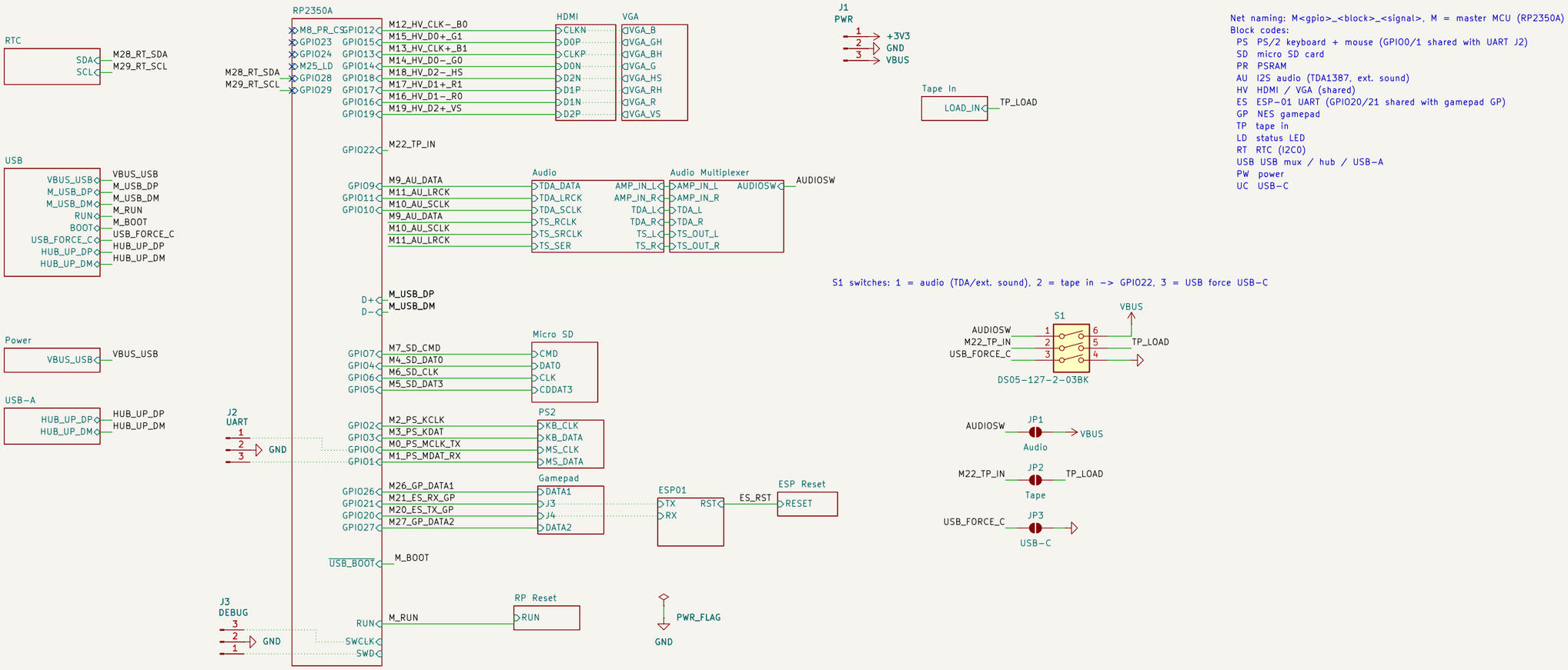


Net naming: M<gpio>_<block>_<signal>, M = master MCU (RP2350A)
Block codes:
PS PS/2 keyboard + mouse (GPIO0/1 shared with UART J2)
SD micro SD card
PR PSRAM
AU I2S audio (TDA1387, ext. sound)
HV HDMI / VGA (shared)
ES ESP-01 UART (GPIO20/21 shared with gamepad GP)
GP NES gamepad
TP tape in
LD status LED
RT RTC (I2C0)
USB USB mux / hub / USB-A
PW power
UC USB-C

S1 switches: 1 = audio (TDA/ext. sound), 2 = tape in -> GPIO22, 3 = USB force USB-C

Shared interfaces (not simultaneous):
Remove ESP-01 before gamepad firmware drives GPIO21: ESP TX is push-pull.
Disconnect UART J2 when using the PS/2 mouse; connect only HDMI or VGA, not both.
Power: J1 is a rail breakout, not an isolated auxiliary input. No USB-C current detection.
Size the 5 V source for board + peripherals, within F1; a PC USB port may not support the full load.

Net names: <M><gpio>_<block>_<signal> (M = RP2350A)
https://github.com/rh1tech/frank

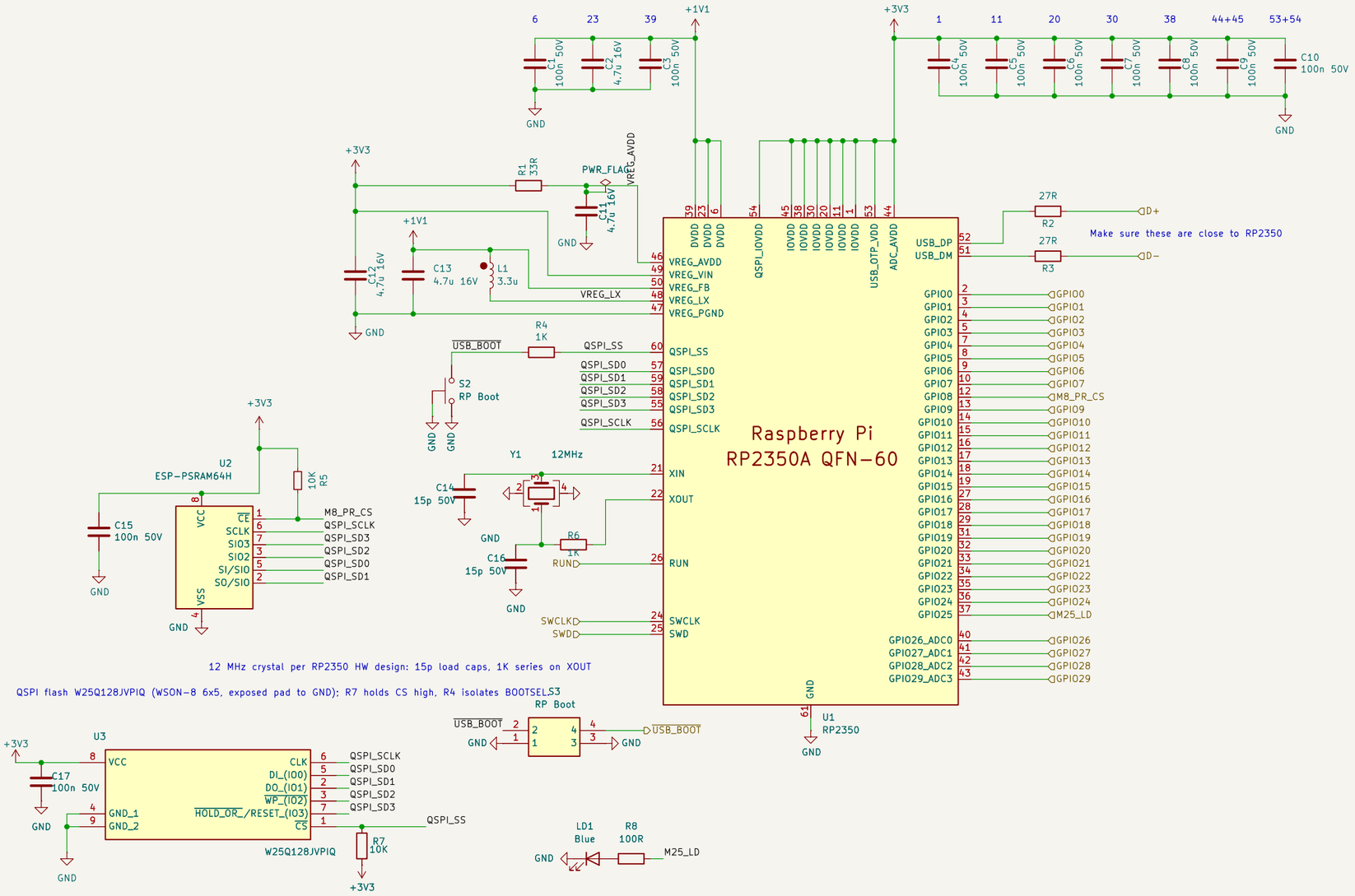
Mikhail Matveev

Sheet: /
File: frank_mini.kicad_sch

Title: FRANK Mini

Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 1/19

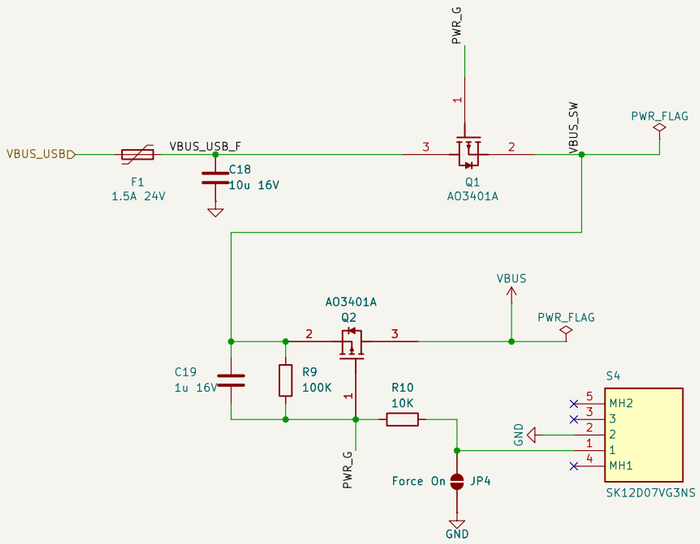
PCB REVIEW: C12 (CIN) and C13 (COUT) are on B.Cu; move beside U1/L1 on F.Cu before fabrication.
Follow RP2350 datasheet 6.3.8: short local power loop, paired GND vias, COUT-to-FB sense, marked L1 orientation.
Keep top copper clear under L1 and layer-2 copper clear under L1/LX; the keepout alone does not fix the capacitor layout.



Block code: M (RP2350A), PR (PSRAM), LD
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /RP2350A/
File: rp2350a.kicad_sch

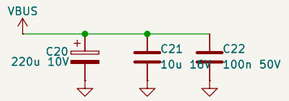
Title: FRANK Mini

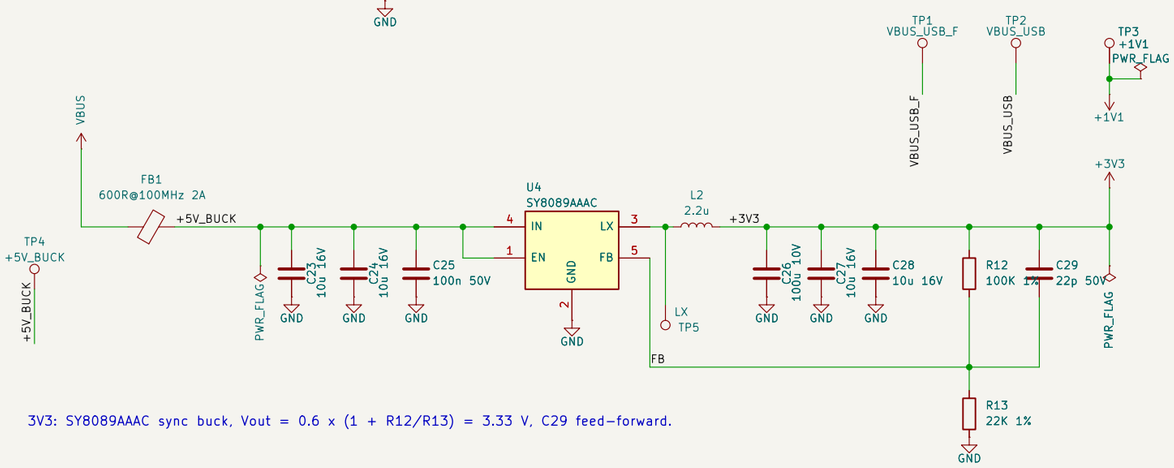
Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 2/19



USB-C 5 V in: F1 polyfuse, Q1/Q2 back-to-back soft switch.
Reverse blocking is OFF-state only; do not feed J1/VBUS while USB-C is connected.

S4 pulls PWR_G low; gate RC tau ~= (R9 || R10) x C19 = 9.1 ms.
R9 returns the gates to the sources on turn-off; JP4 Force On bypasses S4.
This is not a regulated current limiter; measure USB hot-plug inrush with the full load.





3V3: SY8089AAAC sync buck, Vout = 0.6 x (1 + R12/R13) = 3.33 V, C29 feed-forward.

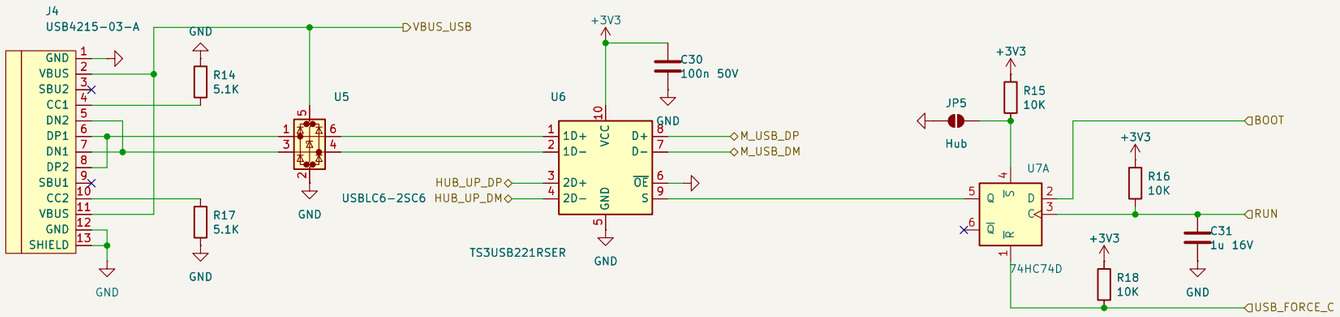
Block code: PW (power)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Power/
File: power.kicad_sch

Title: FRANK Mini

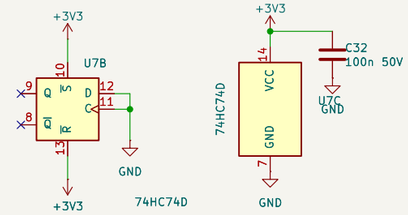
Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 3/19

USB-C: CC 5.1K = sink; USBLC6 ESD near the connector. Series 27R is at the RP2350 (R2/R3).
TS3USB221 mux: S = L -> USB-C (BOOTSEL / device), S = H -> hub (RP is USB host for keyboard / mouse).



BOOTSEL latch U7A: on the RUN rising edge it stores the BOOTSEL button: held -> Q = L -> USB-C, released -> Q = H -> hub.

R16/C31 delay RUN (10 ms RC). Verify the stocked 74HC74D vendor's clock slew limits.
74HC74D is not 74HCS74; steady D alone does not qualify a slow clock input.
JP5 forces hub (~S): S1-3 / JP3 force USB-C (~R). Never enable both overrides.



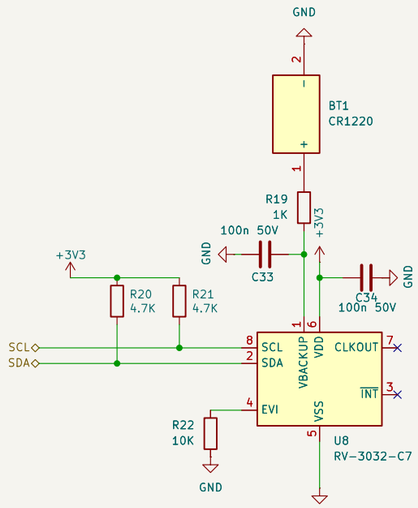
U7B unused: inputs tied off.

Block code: UC (USB-C), USB (mux, BOOTSEL latch)
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /USB/
File: usb.kicad_sch

Title: FRANK Mini

Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 4/19

RV-3032-C7: backup switchover is DISABLED on delivery. Configure once with valid VDD:
Set EERD=1 (10h bit 2), then wait for EEbusy=0 (0Eh bit 2) before changing configuration.
Write C0h RAM mirror: BSM=10 (LSM), TCM=00 (charger OFF for CR1220): optionally NCLKE=1.
Persist with EECMD (3Fh)=11h; wait at least 10 ms, then poll EEbusy=0 and check EEF.
Finally clear EERD=0. Do not rewrite EEPROM on every boot.



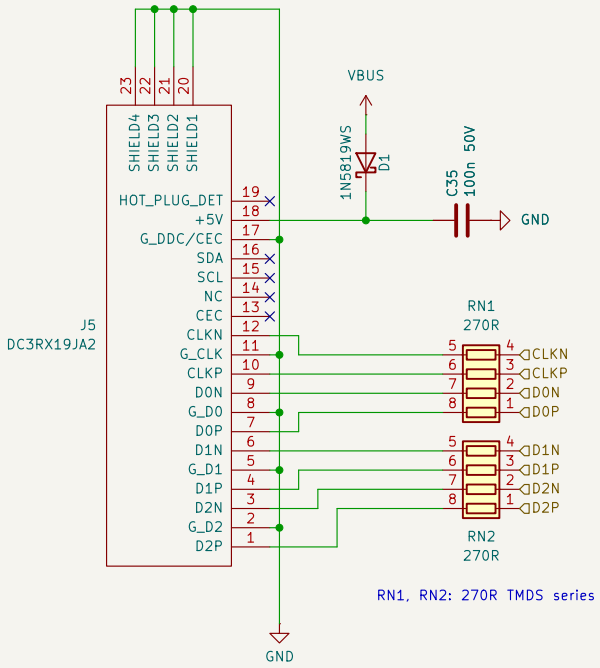
4.7K I2C pull-ups R20/R21 (I2C0: GPIO28 SDA / GPIO29 SCL)

Block code: RT
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /RTC/
File: rtc.kicad_sch

Title: FRANK Mini

Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 5/19

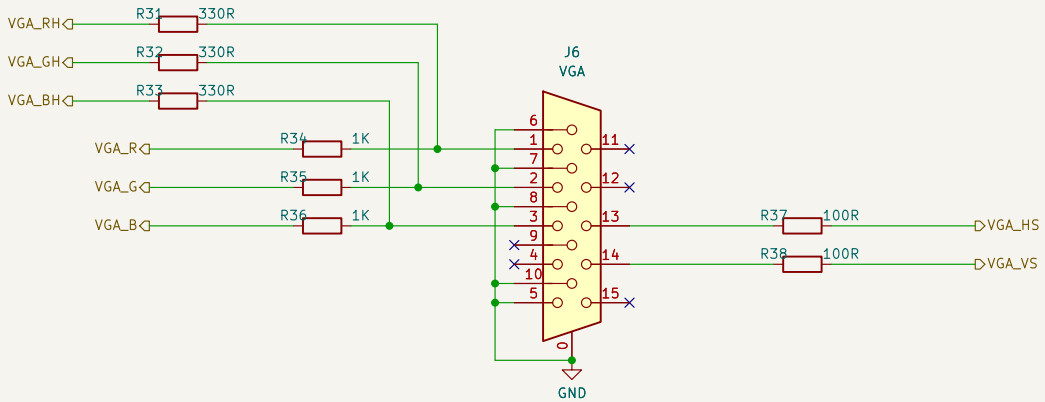


Block code: HV
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /HDMI/
File: hdmi.kicad_sch

Title: FRANK Mini

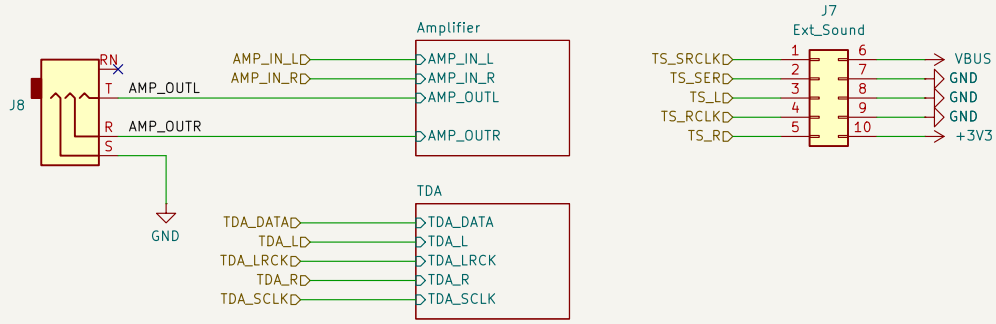
Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 6/19



Block code: HV
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /VGA/
File: vga.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 7/19

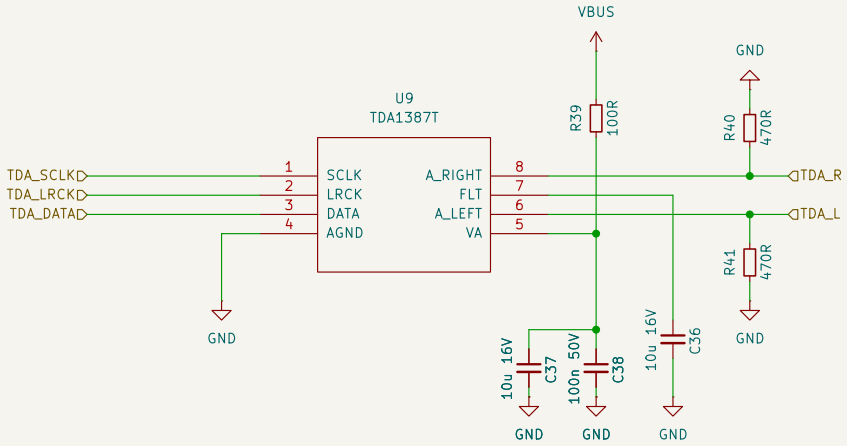


Block code: AU
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio/
File: audio.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 8/19



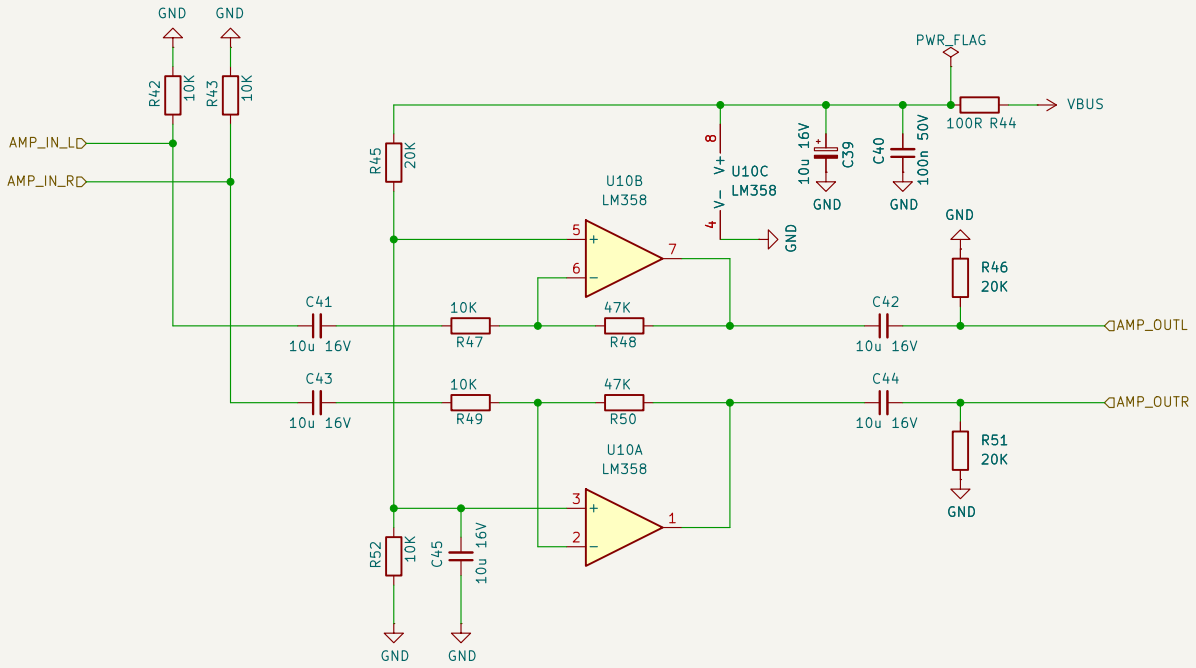
Block code: AU
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio/TDA/
File: tda.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 9/19

LM358 is not rail-to-rail: R45/R52 bias at V+/3 to preserve positive output headroom.
Line output only (>=10K load); not a headphone or speaker driver.

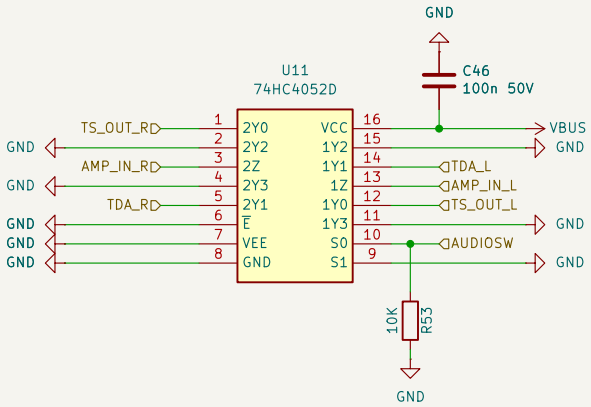


Block code: AU
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio/Amplifier/
File: amp.kicad_sch

Title: FRANK Mini

Size: A4 | Date: 2026-11-05 | **Rev: 1.00**
KiCad E.D.A. 10.0.1 | Id: 10/19

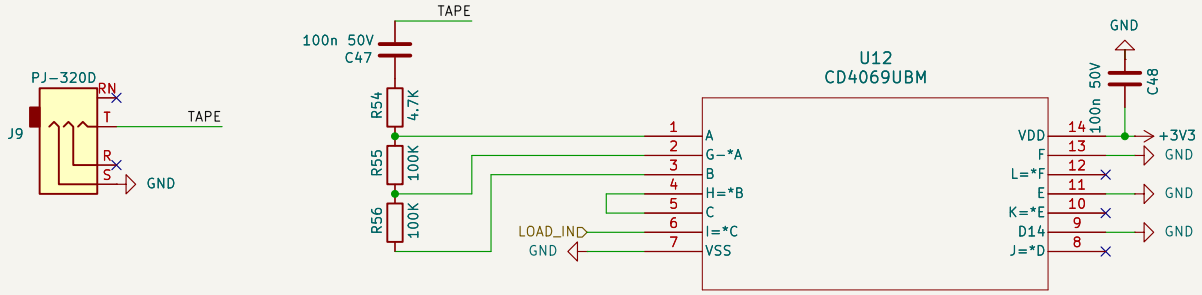


Block code: AU
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Audio Multiplexer/
File: audio_multiplexer.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 11/19



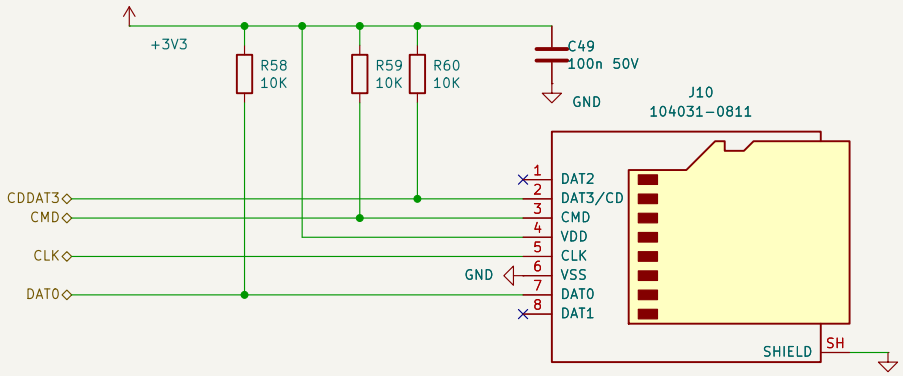
Block code: TP
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Tape In/
File: tape.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 12/19

10K pull-ups on CS, CMD, DAT0 (SPI); DAT1/DAT2 unused; C49 at the socket



Block code: SD
https://github.com/rh1tech/frank
Mikhail Matveev

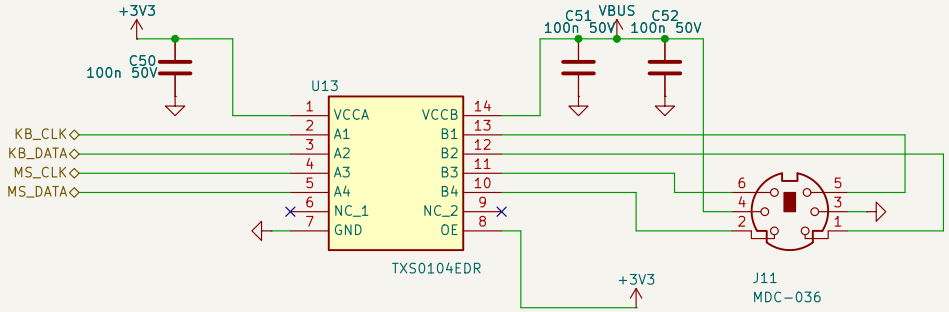
Sheet: /Micro SD/
File: sd.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 13/19

PS/2 keyboard + mouse on one Mini-DIN (J11): TXS0104E 3V3 <-> 5V, OE tied high.

C50 VCCA, C51 VCCB at U13; C52 at the connector VBUS pin. U13 has internal 10K pull-ups on both sides.

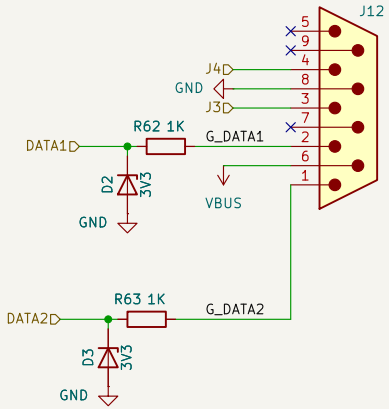


Block code: PS
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /PS2/
File: ps2.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 14/19



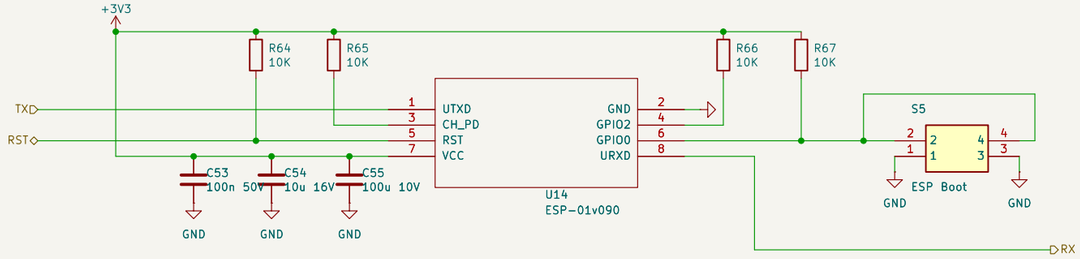
Block code: GP
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /Gamepad/
File: gamepad.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 15/19

ESP-01 socket: 10K pull-ups keep EN/RST/GPIO0/GPIO2 high for normal boot; hold S5 "ESP Boot" + press ESP Reset to flash it.
UART: TX = ESP UTXD -> RP GPIO21, RX = ESP URXD <- RP GPIO20 (shared with the gamepad port).

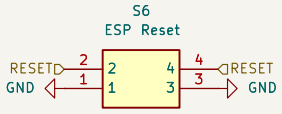


Block code: ES
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /ESP01/
File: esp01s.kicad_sch

Title: FRANK Mini

Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 16/19

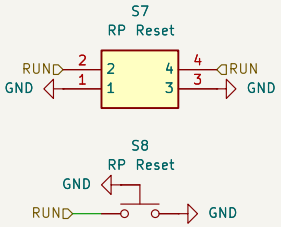


Block code: ES
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /ESP Reset/
File: reset_esp.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 17/19



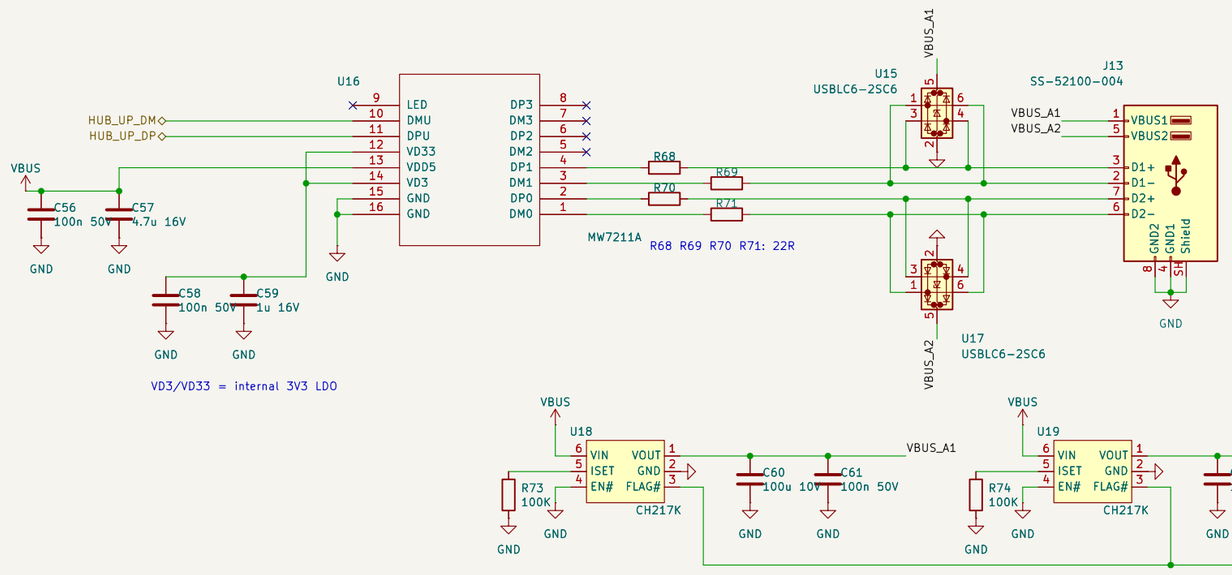
Block code: M (RUN)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /RP Reset/
File: reset_rp.kicad_sch

Title: FRANK Mini

Size: A4	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 18/19

USB hub MW7211A: ports 1/2 -> stacked USB-A J13 (22R series, USBLC6 ESD near the connector), ports 3/4 unused.



USB-A port power: CH217K per port, ISET 100K -> 0.6 A limit (Iset = 60K/Rset). Red LD3 = over-current on either port.

Block code: USB (hub, USB-A ports)
https://github.com/rh1tech/frank
Mikhail Matveev

Sheet: /USB-A/
File: usb_a.kicad_sch

Title: FRANK Mini

Size: A3	Date: 2026-11-05	**Rev: 1.00**
KiCad E.D.A. 10.0.1		Id: 19/19